A Unified Dual-Helix Framework for Geometrically-Guided Pollard's Rho

Cyclic Feature Embeddings, Quadrance Oracles, and Exact Certification

BioChain AI — SHD-CCP Architecture Research

Consolidated Technical Report

Abstract

This report consolidates five source manuscripts on the Schism Dual-Helix / Continuous-Certification Protocol (SHD-CCP) into a single technical framework for accelerating the integer-factorization routine Pollard's Rho. All constructions, definitions, and results of the source manuscripts are preserved. The framework rests on one governing principle—*geometry helps search; arithmetic proves correctness*—realized as a Dual-Helix architecture: a continuous, heuristic *geometric oracle* (Helix A) that ranks candidate collision indices, and a discrete, *exact certifier* (Helix B) that reconstructs true modular residues and applies a classical GCD test. We show that this architecture is *unconditionally exact* whenever it reports a factor, and that its state-reconstruction cost is $\mathcal{O}(\sqrt{T})$ via standard checkpointing. Its *acceleration* over classical baselines is not proven; it is contingent on a single empirical premise—the *Oracle Bias Hypothesis* $\beta > 1$—for which no evidence is currently supplied. We integrate the harmonic (torus), quaternionic (S^3), and hyperbolic feature embeddings; the path-length, Euclidean-quadrance, and rational-hyperbolic-quadrance scores; the empirical linear surrogate and the heuristic feature combinator; and the Fokker–Planck / FP4 candidate sampler. Two source claims are explicitly re-scoped: the polylogarithmic *jump-ahead* of the operator-embedding manuscript is presented as an open problem (and is shown to contradict a standard cryptographic hardness assumption in the $c = 0$ case), and the $\mathcal{O}(\sqrt{N \log N})$ reconstruction bound is presented as unsupported and superseded by the honest $\mathcal{O}(\sqrt{T})$ analysis. Every quantitative claim is annotated with a status tag; see Table 1.

Editorial Note. This is a *synthesis* document. Text originating from the source manuscripts is reproduced faithfully. Text added to link the manuscripts, to reconcile conflicts between them, or to record a critique is marked as an **Editorial Note**, an **Obstruction**, or with an inline status tag: **[PROVEN]**, **[HEURISTIC]**, **[CONJECTURAL]**, **[OPEN]**, **[SUPERSEDED]**. The status vocabulary is: *Proven* (a correct mathematical statement with a valid proof); *Heuristic* (an engineering device offered without a theorem, and explicitly not claimed as one); *Conjectural* (an unproven premise the framework depends on); *Open* (a target that would require new mathematics and is not achieved here); *Superseded* (a source claim replaced by a stronger or more defensible statement elsewhere in this report).

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1 Introduction: The Dual-Helix Governance Model

Given a composite integer N (equivalently n; the source manuscripts use both), classical Pollard's Rho seeks a nontrivial prime factor p by iterating the quadratic modular recurrence

$$x_{t+1} \equiv x_t^2 + c \pmod{N}, \qquad f(x) = x^2 + c. \tag{1}$$

A collision $x_i \equiv x_j \pmod{p}$ implies $p \mid \gcd(x_i - x_j, N)$; if

$$1 < d = \gcd(x_i - x_j, N) < N, \tag{2}$$

a nontrivial factor has been found. Classical implementations locate such collisions via Floyd's or Brent's cycle detection.

The SHD-CCP architecture is an *exact-verification wrapper* around this routine that decouples heuristic candidate generation from exact certification:

- **Helix A (Heuristic Oracle).** A massively parallel, low-precision, continuous topological space. Discrete modular states are embedded into a geometric manifold; a scalar feature score ranks candidate index pairs (i, j) by geometric proximity. Helix A is a candidate *generator* only; its outputs are strictly untrusted.
- **Helix B (Authoritative Certification).** A rigid discrete integer framework. It reconstructs the exact residues $x_i, x_j \in \mathbb{Z}/N\mathbb{Z}$ for the top-ranked candidates and applies the classical test (2).

Editorial Note. The five source manuscripts describe *variants of Helix A* (different embeddings and scores) sharing *one Helix B* (GCD certification plus state reconstruction). The logical spine of this report is therefore: (i) establish why exact algebraic acceleration is unavailable (§2); (ii) motivate the heuristic-oracle detour (§1); (iii) catalogue the oracle constructions (§3); (iv) state the exact and honestly-bounded certifier (§4); (v) isolate the single conjecture on which all acceleration depends (§5); and (vi) give the protocol that would test it (§6).

2 The Algebraic Obstruction to Exact Acceleration

The natural way to accelerate (1) would be an *exact jump-ahead*: computing $f^T(x)$ (the T-fold composition) in $O(\log T)$ operations, as one does for elliptic-curve scalar multiplication via double-and-add, or for linear recurrences via matrix exponentiation. Several independent obstructions rule this out for the generic quadratic map.

2.1 Infinite-dimensionality of exact linearization

To linearize the nonlinear map (1) one seeks a lift Φ and a matrix $\mathbf{M}_c$ with

$$\Phi(x^2 + c) = \mathbf{M}_c \cdot \Phi(x). \tag{3}$$

Under Carleman linearization or the Koopman operator, this requires an *infinite-dimensional* matrix, because the degree of x continuously doubles under squaring, generating the tower $1, x, x^2, x^4, \ldots$. Truncating the infinite matrix yields only an *approximation*, which is useless for an exact algorithm evaluated modulo N.

2.2 The finite-dimension pigeonhole obstruction

A fixed finite (e.g. 64×64) linear embedding cannot universally embed the exact chronologies of (1) for arbitrary N. Because $\mathbb{Z}/N\mathbb{Z}$ contains N states, exact point-to-point chronological recovery requires an N-dimensional space (e.g. a one-hot embedding $\Phi(x) = e_x \in \mathbb{R}^N$). Forcing N states into a static 64-dimensional container triggers the pigeonhole principle: $\Phi(a) = \Phi(b)$ for some $a \neq b$. Once two states collide, their future trajectories cannot be differentiated, destroying exact modular recovery. Hence the 64×64 Toda/Kogame geometry $\mathcal{H}_{64}$ cannot act as the global state container.

2.3 Geometric framing via the Triple Quad Formula

The operator-embedding manuscript reframes squaring geometrically using *Quadrance* (squared distance). The Triple Quad Formula (TQF) states the exact condition for three collinear points,

$$(Q_1 + Q_2 + Q_3)^2 = 2\,(Q_1^2 + Q_2^2 + Q_3^2), \tag{4}$$

and the manuscript proposes to "prune Q_2" so that any scalar growth registering as flat geometrically frustrates unbounded expansion, redirecting the (otherwise infinite) algebraic tower into a bounded geometric shape.

Editorial Note. Equation (4) is a correct classical identity of rational trigonometry. However, "pruning Q_2" is a heuristic narrative, not a proof that the resulting bounded object retains enough information to recover $f^T(x)$ exactly. The pigeonhole obstruction above shows that no bounded-dimensional object can retain that information globally. The manuscript itself acknowledges this and pivots to a "generative kernel" interpretation (§2.6); we track that pivot honestly there.

2.4 No addition law for the generic quadratic map

For elliptic curves, points form an abelian group with a closed-form rational addition law $P+Q$, so iP is computed in $O(\log i)$ by double-and-add. For the pseudo-random walk (1) this luxury is absent. To leap from step i to step $i + j$ without computing intermediate states, one would need an addition formula

$$f^{(i+j)}(x) = \mathrm{Combine}\big(f^{(i)}(x),\, f^{(j)}(x),\, c,\, N\big) \pmod N. \tag{5}$$

No known efficiently computable such Combine exists for generic c, and the SHD-CCP architecture does not assume one exists.

2.5 Solvable special cases

Two boundary values of c admit exact structure:

- $c = 0$: $f^{(i)}(x) \equiv x^{2^i} \pmod N$. Iteration is exact exponentiation (repeated squaring of the *argument*).
- $c = -2$: under the substitution $x = z + z^{-1}$, iteration corresponds to $z \mapsto z^2$, giving Chebyshev-type identities $T_m(T_n(x)) = T_{mn}(x)$.

These cases are correct and useful for testing, but they are measure-zero in c and do not extend to the generic recurrence.

2.6 The jump-ahead as an open problem

The operator-embedding manuscript proposes to escape the pigeonhole obstruction by treating the 64-vertex geometry not as a *container* but as a *generative kernel*: a local rule of growth $\mathbf{M}_{64}$ whose Sierpinski-self-similar folding weaves an expanding toroidal accumulator $\mathcal{M}_{\text{torus}}$ capable of holding N states. To yield a sub-linear jump-ahead, four conditions are posited:

(I) An injective encoder $E_N : \mathbb{Z}/N\mathbb{Z} \to \mathcal{M}_{\text{torus}}$.

(II) A local generator $G_{\mathbf{M}_{64}} : \mathcal{M}_{\text{torus}} \to \mathcal{M}_{\text{torus}}$.

(III) An exact decoder $D_N : \mathcal{M}_{\text{torus}} \to \mathbb{Z}/N\mathbb{Z}$.

(IV) Exact equivariance:

$$D_N\big(G^T_{\mathbf{M}_{64}}(E_N(x))\big) = f^T(x) \pmod N. \tag{6}$$

The manuscript then asserts a "Quadrance Scaling Rule" making the fold isomorphic to an associative algebraic operation, hence a semigroup relation and a repeated-squaring shortcut,

$$G^{a+b} = G^a \circ G^b, \qquad G^T = G^{2^{b_1}} \circ G^{2^{b_2}} \circ \cdots, \tag{7}$$

claimed to compute the step-T state in $O(\text{poly}(\log T, \log N))$.

Open Problem 2.1 (Exact sub-linear jump-ahead for the quadratic map)**. [OPEN]** Construct explicit maps $(E_N, G_{\mathbf{M}_{64}}, D_N)$ satisfying (I)–(IV) such that a single application of G has cost bounded independently of T, thereby realizing (6) in $O(\text{poly}(\log T, \log N))$.

Editorial Note. Conditions (I)–(IV) are the *correct* formal statement of what a jump-ahead requires; this is a genuine contribution of the source manuscript. However, the manuscript does not *construct* maps satisfying them—it asserts their existence via "Quadrance." The semigroup relation (7) is trivially true for *any* iterated map (function composition is associative), but associativity alone does not yield a cheap G^{2^k}. Matrix/group exponentiation is fast because each squaring acts on a *fixed-size* object at bounded cost; the pigeonhole and infinite-dimensionality obstructions above are exactly the statement that no such fixed-size, exactly-composable object exists for the generic quadratic map. Problem 2.1 is therefore not solved here.

Obstruction (The $c = 0$ case is at least as hard as factoring N). Consider Problem 2.1 at $c = 0$, where $f^T(x) \equiv x^{2^T} \pmod N$. A polylogarithmic method for computing $x^{2^T} \bmod N$ from (x, T, N) *without* the factorization of N would violate the repeated-squaring / time-lock assumption (Rivest–Shamir–Wagner): reducing the exponent 2^T modulo $\lambda(N) = \text{lcm}(p-1, q-1)$ requires knowing $\lambda(N)$, i.e. the factorization of N. Since recovering that factorization is the very goal of the algorithm, a working jump-ahead in this case would be *circular*—it would presuppose the answer. This does not merely leave Problem 2.1 unproven; it makes the claimed capability contradict a standard cryptographic hardness assumption in the one sub-case that can be checked exactly.

Editorial Note. **Consequence for the framework.** Because Problem 2.1 is unsolved, the working architecture does *not* use a jump-ahead. The remaining four manuscripts are built on the opposite premise (no shortcut exists), which is precisely why they require a heuristic oracle plus $\mathcal{O}(\sqrt{T})$ checkpointed reconstruction (§4). The operator-embedding manuscript is thus positioned here as *motivation and open problem*, not as a delivered acceleration mechanism. Its conclusion—that G^{2^k} can be computed in $O(\mathrm{poly}(\log T, \log N))$—is reclassified **[OPEN]**.

3 Helix A: The c-Parameterized Geometric Feature Space

Since exact acceleration is unavailable, Helix A pursues *heuristic feature correlation*: embed the orbit into a continuous manifold, compress each state to a scalar score, and use those scores to bias candidate selection. This section catalogues the embeddings and scores from all five manuscripts and reconciles their conflicts.

3.1 Cyclic feature maps and the modular-invariance requirement

A continuous feature attached to the parameter c must be well-defined on the residue class $c \in \mathbb{Z}/N\mathbb{Z}$. A fractional embedding c/N fails this: since $c \equiv c + kN \pmod N$, the value c/N depends on the chosen integer representative and is *not* modular invariant. To be rigorous the embedding must be perfectly periodic.

Definition 3.1 (Cyclic (harmonic) feature map)**.** Define the continuous periodic feature

$$\psi(c) = \cos\left(\frac{2\pi c}{N}\right). \tag{8}$$

Then $c \equiv c' \pmod N \Rightarrow \psi(c) = \psi(c')$. The map is not injective on $\mathbb{Z}/N\mathbb{Z}$ (since $\psi(c) = \psi(N-c)$); it intentionally compresses c, losing directional information. A fully injective encoding uses the complete root of unity $\Psi(c) = e^{2\pi i c/N}$, with $\psi(c) = \mathrm{Re}\,\Psi(c)$.

3.2 The metric deformation: reconciling the two proposals

Two source manuscripts deform a 2D metric tensor on a flat torus by c. They disagree, and one is invalid by the criterion of §3.1.

Proposal A (harmonic, retained).

Definition 3.2 (Positive-definite harmonic metric)**.** Let $\mathcal{M}_c$ be a flat torus $\mathbb{T}^2 = \mathbb{R}^2/\mathbb{Z}^2$. With coupling constant $|\lambda| < 1$,

$$g_{\mu\nu}(c) = \begin{pmatrix} 1 & \lambda\psi(c) \\ \lambda\psi(c) & 1 \end{pmatrix}. \tag{9}$$

Proposition 3.3 (Strict positive-definiteness)**.** ***[PROVEN]*** $\det g_{\mu\nu}(c) = 1 - \lambda^2\psi(c)^2$. *Since* $|\lambda| < 1$ *and* $|\psi(c)| \le 1$, $\det g_{\mu\nu}(c) > 0$ *for all* c*; the eigenvalues* $\mu_{1,2}(c) = 1 \pm \lambda\psi(c)$ *are strictly positive; the metric is strictly positive definite. For each fixed* c *it is a constant flat metric—deforming inner products on tangent spaces without introducing coordinate-dependent curvature.*

Proposal B (golden shear, corrected). The geodesic-embedding manuscript instead proposes, with the golden ratio φ,

$$g_{\mu\nu}(c) = \begin{pmatrix} 1 & \frac{c\varphi}{N} \\ \frac{c\varphi}{N} & \varphi^2 \end{pmatrix}, \tag{10}$$

the fractional shear $c\varphi/N$ "physically stretching" the manifold.

Editorial Note. The shear term $c\varphi/N$ in (10) is a fractional embedding in c/N and therefore *not modular invariant* by §3.1: the same residue class yields different geometries under different integer representatives of c. We reconcile the two proposals by adopting (9), i.e. replacing the offending term $c\varphi/N \mapsto \lambda\psi(c) = \lambda\cos(2\pi c/N)$. Metric (10) is retained here only as the historical proposal; it is reclassified **[SUPERSEDED]** in favor of (9).

3.3 State embeddings into the manifold

The manuscripts supply three concrete embeddings of the discrete orbit; all are retained as members of one family of heuristic feature maps $E_{N,c} : \mathbb{Z}/N\mathbb{Z} \to \mathcal{M}_c$.

Definition 3.4 (Flat-torus embedding)**.**

$$E_{N,c}(x_t) = \left(\frac{x_t}{N},\ \frac{x_t^2 + c \bmod N}{N}\right) \in \mathbb{T}^2. \tag{11}$$

Consecutive images are joined by shortest-lift displacements $\Delta_t = (\Delta u_t, \Delta v_t) \in [-\frac{1}{2}, \frac{1}{2}]^2$, and γ_c denotes the piecewise-linear interpolation through the lifted points.

Definition 3.5 ($S^3 \subset \mathbb{R}^4$ toroidal embedding)**.** With $r_c^2 + s_c^2 = 1$,

$$E_{N,c}(x) = \left(r_c \cos\tfrac{2\pi x}{N},\ r_c \sin\tfrac{2\pi x}{N},\ s_c \cos\tfrac{2\pi(x^2+c)}{N},\ s_c \sin\tfrac{2\pi(x^2+c)}{N}\right), \tag{12}$$

a flat torus in $S^3 \subset \mathbb{R}^4$ inheriting the ambient Euclidean metric.

Definition 3.6 (Quaternionic delta encoding and rational Lorentz lift)**.** The step x_k is encoded as a unit quaternion rotation with phase proportional to the residue,

$$\Delta q_k = \cos(\theta x_k) + \mathbf{u}\,\sin(\theta x_k), \qquad \theta = \frac{2\pi}{720}, \tag{13}$$

where $\mathbf{u}$ is a unit vector in $\mathbb{R}^3$ and θ obeys the 720° spinor cyclic constraint. The state $q_i \in S^3 \cap \mathbb{Q}^4$ is the ordered product of prior deltas, lifted to the hyperboloid $\mathbb{H}^3$ by the rational Lorentz lift

$$E_n(x_i) = \tilde{q}_i = \left(\frac{1}{w_i}, \frac{x_i}{w_i}, \frac{y_i}{w_i}, \frac{z_i}{w_i}\right). \tag{14}$$

3.4 Feature scores

Definition 3.7 (Path-length functional)**.** The score of an orbit after i steps is the arc length of the interpolated trajectory (a finite sum for the piecewise-linear path):

$$\mathcal{L}_c(i) = \sum_{t=0}^{i-1} \sqrt{\Delta_t^\top\, g(c)\, \Delta_t} = \sum_{t=0}^{i-1} \sqrt{\Delta u_t^2 + 2\lambda\psi(c)\,\Delta u_t \Delta v_t + \Delta v_t^2}. \tag{15}$$

Definition 3.8 (Euclidean quadrance score)**.**

$$\chi_c(i) = \left\|E_{N,c}(x_i) - E_{N,c}(x_0)\right\|_2^2, \qquad 0 \le \chi_c(i) \le 4. \tag{16}$$

Collapsing the 4D state to one nonnegative scalar intentionally discards angular and directional information; χ_c is a rough *radial* feature.

Definition 3.9 (Rational hyperbolic quadrance)**.** For the Lorentz-lifted states,

$$Q_h(i,j) = \langle \tilde{q}_i, \tilde{q}_j \rangle_{\mathbb{L}}^2 - 1, \tag{17}$$

designed to correlate heuristically with the modular difference structure $x_i - x_j \pmod n$; candidate pairs are prioritized where $Q_h(i,j) \to 0$.

Definition 3.10 (Golden geodesic-compression score)**.**

$$\chi_c(i) = \mathcal{Q}_{\text{seed}} \cdot \left[\frac{\varphi^{2\lfloor \log_\varphi i \rfloor}}{\sqrt{1 - \left(\frac{c\varphi}{N}\right)^2}} \right] \times \operatorname{Tr}\left(\mathbf{M}_{64}^{\otimes \lfloor \log_\varphi i \rfloor} \right). \tag{18}$$

Remark 3.11 (Status of the scores). All scores above are heuristic. Every manuscript is explicit that the score is *not* the modular iterate: $\chi_c(i) \not\equiv f^i(x) \pmod N$. Scores (15), (16), (17) are well-defined and computable; they are marked **[HEURISTIC]**. Score (18) is retained verbatim but flagged: it depends on the non-invariant term $c\varphi/N$ (see §3.2) and combines a $\log_\varphi$-indexed golden power with a Kronecker-power trace of $\mathbf{M}_{64}$ without a derivation connecting it to orbit structure; it is marked **[HEURISTIC]** with the caveat that it is not shown to carry collision-relevant signal.

3.5 The evaluation obstruction and the empirical surrogate

Although $\mathcal{L}_c(i)$ is a finite sum, evaluating it for many candidate indices requires the corresponding orbit segments, which requires sequential modular iteration—defeating the point of a low-cost ranking layer. The exact length is therefore replaced by a fitted surrogate.

Heuristic Assumption 3.12 (Statistical linearization)**.** **[HEURISTIC]** The segment lengths $\ell_t(c)$ behave as weakly dependent samples from a stationary distribution with finite mean m_c, so that by an empirical law of large numbers $\mathcal{L}_c(i)/i \to m_c$ as $i \to \infty$.

Under Assumption 3.12, the path-length is approximated by an empirical linear regression

$$\widetilde{\mathcal{L}}_c(i) \approx \alpha(c)\, i + \beta(c), \tag{19}$$

with $\alpha(c), \beta(c)$ fitted from observed orbit data.

Remark 3.13. Model (19) is a feature-engineering heuristic, not a theorem of the orbit; it yields a constant-time scalar per candidate. Because $\Delta\widetilde{\mathcal{L}}_c(i,j) = |\alpha(c)(i-j)|$, a purely linear model predominantly ranks *nearby* indices; a more expressive candidate generator evaluates pairwise features

$$\Phi_c(i,j) = F\big(i, j, \widetilde{\mathcal{L}}_c(i), \widetilde{\mathcal{L}}_c(j), \psi(c)\big) \tag{20}$$

to emit a probability distribution $\mathcal{D}_\chi$ over correlated geometric pairs.

3.6 The heuristic feature combinator and drift

Absent an exact addition law (5), the oracle uses an *approximately* compositional rule modeled on a Euclidean quadrance expansion.

Definition 3.14 (Feature combinator)**.** **[HEURISTIC]**

$$\chi_c(i) *_c \chi_c(j) = \chi_c(i) + \chi_c(j) + 2\,\rho_c\big(\chi_c(i), \chi_c(j)\big)\sqrt{\chi_c(i)\,\chi_c(j)}, \qquad -1 \le \rho_c \le 1, \tag{21}$$

where ρ_c is a correlation parameter learned from localized orbit data. The rule preserves non-negativity and acts as a predictive radial model for feature concatenation, estimating proximity without generic associativity.

Definition 3.15 (Normalized feature drift)**.**

$$\delta_{i,j} = \frac{\big|\chi_c(i+j) - \big(\chi_c(i) *_c \chi_c(j)\big)\big|}{1 + \big|\chi_c(i+j)\big|}. \tag{22}$$

As $\delta_{i,j}$ grows, the predictive advantage of the combinator degrades and the oracle's sampling distribution reverts toward uniformity.

3.7 The Fokker–Planck / FP4 candidate sampler

The oracle's probability mass is evolved by a continuous diffusion whose tensor is steered by a geometric feature.

Definition 3.16 (Fokker–Planck prime sieve)**.** **[HEURISTIC]**

$$\frac{\partial \rho}{\partial t} = \nabla \cdot \big(D(Q)\nabla\rho\big) - V(Q)\rho, \qquad D(Q) = \frac{1}{1+\alpha\cdot\chi(Q)}, \tag{23}$$

with $\chi(Q)$ the TQF collinearity feature. Probability gradients are executed in 4-bit floating point (E4M3 NVFP4). A deterministic "Smallest-Three" bounded-error quantization induces an equivalence relation

$$u \sim_{\mathrm{FP4}} v \iff \mathrm{SmallestThree}(u) = \mathrm{SmallestThree}(v), \tag{24}$$

connecting states whose rounded FP4 feature vectors agree and forming a quantization-induced clique complex. Projecting the sequence onto the irrational Golden Strassen–Clifford Toroid ($r_1/r_2 = \varphi$) prior to quantization is intended to heuristically decorrelate trivial global N-cycles from persistent local structure.

Editorial Note. The constructions of §3.6–§3.7 are internally consistent as *feature engineering*. None is accompanied by a theorem relating its output to modular collision probability; each is correctly labeled heuristic by its own manuscript. Their collective purpose is only to *shape* the sampling distribution $\mathcal{D}_\chi$, whose usefulness is the subject of the single conjecture in §5.

4 Helix B: Exact Certification and State Reconstruction

Because every Helix A score is a lossy scalar that cannot recover exact residues, Helix B must reconstruct the true $x_i, x_j \in \mathbb{Z}/N\mathbb{Z}$ and apply the exact test. This layer carries the framework's only unconditional guarantees.

4.1 Exact GCD certification

Theorem 4.1 (Exactness of Dual-Helix certification)**.** ***[PROVEN]*** *Let N be composite and $x_{k+1} \equiv x_k^2 + c \pmod N$. Suppose Helix A outputs candidate indices (i, j). For each pair, Helix B reconstructs the exact residues x_i, x_j via checkpointed procedural seeds and computes $d = \gcd(x_i - x_j,\, N)$. Then every reported factor d satisfying $1 < d < N$ is a mathematically certified nontrivial factor of N, unconditionally and independently of the correctness or resolution of the heuristic oracle.*

Proof. Since $d = \gcd(x_i - x_j, N)$, we have $d \mid N$. The condition $1 < d < N$ excludes the trivial divisors 1 and N, so d is a proper nontrivial factor. The oracle influences only which candidate pairs are proposed (the bias β); it cannot affect the arithmetic value of the GCD. Hence the validity of the reported factor is fully isolated from any topological assumption or FP4 quantization boundary. □

4.2 State reconstruction via structural checkpointing

Retrieving x_i from x_0 by naive iteration is $\mathcal{O}(N)$ in the worst case. A standard memory–time tradeoff removes this.

Theorem 4.2 (Checkpointed reconstruction)**.** ***[PROVEN]*** *Fix a sequence horizon T. Store checkpoints every $L = \lceil\sqrt{T}\rceil$ steps. To reconstruct any x_i with $0 \le i \le T$, restart from the nearest preceding checkpoint $m = \lfloor i/L \rfloor$ and iterate forward at most $L - 1$ steps. This yields*

$$T_{\mathit{reconstruct}}(i) \in \mathcal{O}(\sqrt{T}), \qquad M(T) \in \mathcal{O}(\sqrt{T}). \tag{25}$$

Proof. There are $\lceil T/L \rceil = \mathcal{O}(\sqrt{T})$ checkpoints, giving the memory bound; each reconstruction performs at most $L - 1 = \mathcal{O}(\sqrt{T})$ forward steps. □

Reconciling the reconstruction claims. Two manuscripts instead assert a reconstruction bound $T_{\text{reconstruct}}(i, j) \in \mathcal{O}(\sqrt{N \log N})$ via "SHD-CCP indexing" / "Prime Triplet Indexing" over the generator set

$$\mathcal{I} = \{(p,\, p+2,\, p+6) \mid p \in \mathbb{P}\}, \tag{26}$$

using a 64-bit "Einstein Tile" packet as a deterministic procedural seed.

Editorial Note. The bound $\mathcal{O}(\sqrt{N \log N})$ is stated without derivation, and no mechanism is given by which the prime-triplet set (26) enables jumping within a pseudo-random orbit; (26) is a sparse arithmetic object unrelated to the recurrence's temporal structure. Theorem 4.2 is the defensible statement, with the horizon T (not N) as the governing parameter. We therefore adopt (25) and reclassify the $\mathcal{O}(\sqrt{N \log N})$ claim as **[SUPERSEDED]**. The Einstein-Tile packet and the generator set (26) are retained as engineering descriptions of the checkpoint encoding, not as a proof of sub-linear reconstruction.

5 The Oracle Bias Hypothesis and Conditional Acceleration

All five manuscripts reduce, in the end, to one premise: that geometric proximity in the chosen embedding correlates with a hidden modular collision. We state it once, in unified form.

Heuristic Assumption 5.1 (Oracle Bias Hypothesis)**. [CONJECTURAL]** Let $\Omega_T = \{(i, j) : 0 \le i < j \le T\}$ and let $\mathcal{U} = \text{Unif}(\Omega_T)$ be the uniform baseline. Let $\mathcal{D}_\chi$ be the oracle distribution induced by a feature score. Let $B = \{(i, j) : 1 < \gcd(x_i - x_j, N) < N\}$ be the success event, and $p = \Pr_{\mathcal{U}}(B) > 0$ the baseline success probability. The oracle exhibits a positive bias $\beta > 1$ if

$$\Pr_{(i,j)\sim\mathcal{D}_\chi}(B) \;\ge\; \beta \cdot \Pr_{(i,j)\sim\mathcal{U}}(B), \tag{27}$$

which is meaningful only when $\beta p \le 1$.

Editorial Note. Assumption 5.1 is the load-bearing, *unproven* claim of the entire framework. No manuscript supplies evidence for it, and there is no a priori reason a continuous embedding of a pseudo-random sequence should place geometric neighbors in correspondence with collisions modulo an *unknown* prime factor p. Establishing $\beta > 1$ empirically—and quantifying it—is the sole determinant of whether the architecture accelerates anything (§6).

Theorem 5.2 (Conditional acceleration)**. *[PROVEN] (arithmetic), conditional on Assumption 5.1. Let each baseline trial cost $C_{baseline}(T, N)$ and each oracle trial cost at most $C_{oracle}(T) + C_{reconstruct}(T) + C_{gcd}(N)$. Under (27), the oracle method has expected cost bounded by***

$$T_{oracle} \;\le\; \frac{C_{oracle}(T) + C_{reconstruct}(T) + C_{gcd}(N)}{\beta p}. \tag{28}$$

Consequently the oracle outperforms uniform traversal if and only if

$$C_{oracle}(T) + C_{reconstruct}(T) + C_{gcd}(N) \;<\; \beta \cdot C_{baseline}(T, N). \tag{29}$$

Proof. The expected number of oracle trials to hit B is at most $1/\Pr_{\mathcal{D}_\chi}(B) \le 1/(\beta p)$ by (27); multiplying by the per-trial cost gives (28). Comparing (28) against the baseline expected cost C_{baseline}/p yields (29). □

Remark 5.3. The algebra of Theorem 5.2 is elementary and correct; its content is entirely inherited from Assumption 5.1. Because $C_{\text{reconstruct}}(T) \in \mathcal{O}(\sqrt{T})$ (Theorem 4.2) may dominate the per-trial cost, practical acceleration requires β large enough to amortize the reconstruction overhead in (29).

6 Empirical Validation Protocol

The framework's value is an empirical question. We collect the source manuscripts' validation steps into one protocol; passing it is necessary before any acceleration claim may be made.

1. **Path-length linearity.** For semiprime testing moduli $N = pq$, compute the exact finite sum $\mathcal{L}_c(i)$ (15) and fit (19). Report the coefficient of determination R^2 and residual distributions to test Assumption 3.12.

2. **Collision-prediction assessment.** For candidate pairs (i, j) sampled via $\mathcal{D}_\chi$, measure the correlation between the pairwise score $\Phi_c(i, j)$ (equivalently $Q_h(i, j)$) and the occurrence of successful hidden collisions B. This directly estimates β in Assumption 5.1; a value $\beta \leq 1$ falsifies the oracle.

3. **Baseline comparison.** Compare the total expected cost T_{oracle} against classical Floyd's and Brent's cycle detection, verifying inequality (29) operationally rather than assuming it.

7 Status of Claims

Table 1 records the status of every quantitative claim consolidated in this report.

Table 1: Consolidated status of claims.

Claim	**Status**	**Basis / note**
Exact GCD certification, $1 < d < N$ (Thm. 4.1)	**[PROVEN]**	Divisibility; oracle-independent.
$\mathcal{O}(\sqrt{T})$ checkpointed reconstruction (Thm. 4.2)	**[PROVEN]**	Standard memory–time tradeoff.
Metric positive-definiteness for $\lvert\lambda\rvert < 1$ (Prop. 3.3)	**[PROVEN]**	$\det = 1 - \lambda^2\psi^2 > 0$.
Modular invariance of $\psi(c) = \cos(2\pi c/N)$	**[PROVEN]**	Periodicity of cosine.
Special cases $c = 0$ (repeated squaring), $c = -2$ (Chebyshev)	**[PROVEN]**	Exact identities.
Infinite-dimensionality of exact linearization	**[PROVEN]**	Carleman/Koopman degree doubling.
Pigeonhole obstruction for fixed 64×64 embedding	**[PROVEN]**	$N > 64$ forces collisions.
Acceleration inequality (29) (given β)	**[PROVEN]**	Elementary; content is in β.
Harmonic metric / path-length / quadrance scores	**[HEURISTIC]**	Feature maps; no collision theorem.
Empirical linear surrogate (19), combinator (21), drift (22)	**[HEURISTIC]**	Fitted devices, not theorems.
Fokker–Planck / FP4 sampler (23)–(24)	**[HEURISTIC]**	Distribution-shaping only.
Golden geodesic score (18)	**[HEURISTIC]**	Retained; not shown to carry signal.
Oracle Bias $\beta > 1$ (Assump. 5.1)	**[CONJECTURAL]**	Unproven; the load-bearing premise.

Table 1 (continued)

Claim	Status	Basis / note
Polylog jump-ahead / G^{2^k} in $O(\text{poly}(\log T, \log N))$ (Prob. 2.1)	**[OPEN]**	Not constructed; contradicts RSW at $c = 0$.
$\mathcal{O}(\sqrt{N \log N})$ reconstruction via prime-triplet indexing	**[SUPERSEDED]**	Underived; replaced by Thm. 4.2.
Golden shear metric (10) (term $c\varphi/N$)	**[SUPERSEDED]**	Not modular invariant; replaced by (9).

8 Conclusion

By treating the c-parameterized geometric moduli space as a high-speed, continuous *feature space* rather than an exact algebraic container, the SHD-CCP framework offers a principled separation of concerns: geometry proposes, arithmetic disposes. Its guarantees are honest and asymmetric. What is proven stands on solid ground—exact GCD certification (Theorem 4.1), $\mathcal{O}(\sqrt{T})$ reconstruction (Theorem 4.2), the positive-definite flat metric (Proposition 3.3), the algebraic-obstruction results, and the elementary cost inequality. What is heuristic is labeled as such: the embeddings, scores, combinator, and diffusion sampler shape a candidate distribution but carry no theorem of collision. What is conjectural is isolated to a single point—the Oracle Bias Hypothesis $\beta > 1$ (Assumption 5.1)—upon which every acceleration claim depends and for which no evidence is yet supplied. And what remains open is stated plainly: the exact sub-linear jump-ahead (Problem 2.1) is not achieved here and, in the checkable $c = 0$ case, would contradict a standard cryptographic hardness assumption.

The practical thesis is therefore conditional and testable: *if* a feature embedding can be shown empirically to satisfy $\beta > 1$ by a margin large enough to satisfy inequality (29) against the $\mathcal{O}(\sqrt{T})$ reconstruction overhead, *then* the architecture yields a net reduction in factorization time—while, in all cases, every reported factor is unconditionally exact. Resolving that "if" is the task of the validation protocol in §6.

A Notation and Construction Catalogue

For completeness, every symbol and construction consolidated from the source manuscripts is listed below.

Symbol / object	Meaning
$f(x) = x^2 + c$; $x_{t+1} \equiv x_t^2 + c \pmod N$	Quadratic modular recurrence (1).
$\mathbb{Z}/N\mathbb{Z}$, $\mathbb{Z}/n\mathbb{Z}$	Residue ring modulo N (resp. n).
$d = \gcd(x_i - x_j, N)$; $1 < d < N$	Exact certification / success event (2).
$\Phi(x^2 + c) = \mathbf{M}_c \Phi(x)$	Sought linearization (3); infinite-dimensional.
$1, x, x^2, x^4, \ldots$	Degree tower under repeated squaring.
$(Q_1 + Q_2 + Q_3)^2 = 2(Q_1^2 + Q_2^2 + Q_3^2)$	Triple Quad Formula (4).
$\Phi(x) = e_x \in \mathbb{R}^N$; $\Phi(a) = \Phi(b)$	One-hot embedding; pigeonhole collision.

Symbol / object	Meaning
$\mathcal{H}_{64}$, $\mathbf{M}_{64}$, $\mathcal{M}_{\text{torus}}$	Toda/Kogame geometry; generator kernel; toroidal accumulator.
$E_N, G_{\mathbf{M}_{64}}, D_N$; (6)	Encoder / generator / decoder and exact equivariance (jump-ahead conditions I–IV).
$G^{a+b} = G^a \circ G^b$; $G^T = G^{2^{b_1}} \circ \cdots$	Semigroup relation; repeated-squaring target (7).
$f^{(i+j)}(x) = \text{Combine}(\cdots)$	Hypothetical addition law (5); unknown for generic c.
$c = 0:\ f^{(i)}(x) \equiv x^{2^i}$; $c = -2:\ T_m(T_n(x)) = T_{mn}(x)$	Solvable special cases.
$\psi(c) = \cos(2\pi c/N)$; $\Psi(c) = e^{2\pi i c/N}$	Harmonic feature map (8); full injective encoding.
$g_{\mu\nu}(c)$ (9); $\det = 1 - \lambda^2\psi^2$; $\mu_{1,2} = 1 \pm \lambda\psi$	Harmonic flat-torus metric; positive definite for $\lvert\lambda\rvert < 1$.
$g_{\mu\nu}(c)$ (10) with shear $c\varphi/N$	Golden metric (superseded; not modular invariant).
$\mathbb{T}^2 = \mathbb{R}^2/\mathbb{Z}^2$	Flat torus search manifold.
$E_{N,c}(x_t) = (x_t/N, (x_t^2 + c \bmod N)/N)$	Flat-torus embedding (11).
$\Delta_t = (\Delta u_t, \Delta v_t) \in [-\frac{1}{2}, \frac{1}{2}]^2$; γ_c	Shortest-lift displacements; piecewise-linear interpolation.
$E_{N,c}(x) \in S^3 \subset \mathbb{R}^4$, $r_c^2 + s_c^2 = 1$	Toroidal S^3 embedding (12).
$\Delta q_k = \cos(\theta x_k) + \mathbf{u}\sin(\theta x_k)$, $\theta = 2\pi/720$	Quaternionic delta (13); 720° spinor constraint.
$q_i \in S^3 \cap \mathbb{Q}^4$; $\tilde{q}_i = (1/w_i, x_i/w_i, y_i/w_i, z_i/w_i)$	Quaternion state; rational Lorentz lift (14).
$\mathcal{L}_c(i)$ (15); $\mathcal{L}_c(i)/i \to m_c$	Path-length functional; law of large numbers.
$\widetilde{\mathcal{L}}_c(i) \approx \alpha(c)i + \beta(c)$; $\Delta\widetilde{\mathcal{L}}_c(i,j) = \lvert\alpha(c)(i-j)\rvert$	Empirical surrogate (19) and its pairwise difference.
$\chi_c(i) = \lVert E_{N,c}(x_i) - E_{N,c}(x_0)\rVert_2^2 \in [0,4]$	Euclidean quadrance score (16).
$Q_h(i,j) = \langle \tilde{q}_i, \tilde{q}_j \rangle_{\mathbb{L}}^2 - 1$; $Q_h \to 0$	Rational hyperbolic quadrance (17).
$\chi_c(i)$ (18)	Golden geodesic-compression score.
$\chi_c(i) *_c \chi_c(j)$ (21), $-1 \le \rho_c \le 1$	Heuristic feature combinator.
$\delta_{i,j}$ (22)	Normalized feature drift.
$\Phi_c(i,j) = F(\cdots)$ (20); $\mathcal{D}_\chi$	Pairwise feature; oracle candidate distribution.
$\partial_t\rho = \nabla\cdot(D(Q)\nabla\rho) - V(Q)\rho$, $D(Q) = 1/(1+\alpha\chi(Q))$	Fokker–Planck prime sieve (23).
$u \sim_{\text{FP4}} v \iff \text{SmallestThree}(u) = \text{SmallestThree}(v)$	FP4 quantization equivalence (24).
Golden Strassen–Clifford Toroid, $r_1/r_2 = \varphi$	Irrational-ratio projection target.
$L = \lceil\sqrt{T}\rceil$, $m = \lfloor i/L\rfloor$; $T_{\text{reconstruct}}, M(T) \in \mathcal{O}(\sqrt{T})$	Checkpoint schedule and bounds (25).
$\mathcal{I} = \{(p, p+2, p+6) : p \in \mathbb{P}\}$; Einstein Tile	Prime-triplet generator; 64-bit procedural seed.

Symbol / object	Meaning
$\Omega_T = \{(i,j) : 0 \le i < j \le T\}$, $\mathcal{U} = \mathrm{Unif}(\Omega_T)$	Candidate universe; uniform baseline.
B, $p = \Pr_{\mathcal{U}}(B)$, β, $\beta p \le 1$	Success event; baseline probability; bias factor.
$\Pr_{\mathcal{D}_\chi}(B) \ge \beta \Pr_{\mathcal{U}}(B)$ (27)	Oracle Bias Hypothesis.
$C_{\mathrm{baseline}}, C_{\mathrm{oracle}}, C_{\mathrm{reconstruct}}, C_{\mathrm{gcd}}$, T_{oracle} (28)	Cost model and expected-runtime bound.
$\beta C_{\mathrm{baseline}} > C_{\mathrm{oracle}} + C_{\mathrm{reconstruct}} + C_{\mathrm{gcd}}$ (29)	Net-acceleration condition.

Source manuscripts consolidated. (1) *Harmonic Feature Embedding: Cyclic Moduli Spaces for Pollard's Rho Orbits*; (2) *The Golden Strassen–Clifford Geodesic Embedding: A c-Parameterized Geometric Moduli Space*; (3) *The Pseudogroup Law of Quadrance Dynamics: Heuristic Feature Combination for c-Parameterized Orbits*; (4) *Finite-Dimensional Operator Embedding via TQF Sierpinski Emergence*; (5) *Topological Optimization of Pollard's Rho via Quaternionic Lattices and Soliton Collapse Manifolds.*